

What do ICML 2025 and SEEQA 2025 have in common?

3 of 5: From "Acts of God" to Acts of Data: Rethinking Catastrophe Insurance



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Merton College Chapel (Poster Sessions). Thoughts on Quantum Technology Based on Interpretations at SEEQA – A Quantum Conference Hosted at University.

An insurance term that has always sent shivers down my spine: “It was an act of God.”

The phrase implies events so overwhelming—floods, hurricanes, wildfires—that no preparation or model could account for them. It carries a sense of inevitability, as if humans and institutions alike must simply absorb the

Yet today, technology is challenging that fatalism. Satellites now track fires in near real time, AI can confirm wildfire damage within minutes, and quantum algorithms are emerging that may one day simulate catastrophe risks with unprecedented accuracy. What was once considered unusu

“divine chaos” is being pulled into the realm of data, models, and—crucially—governance.

Earth Observation: Making the Invisible Visible

The convergence of Earth Observation (EO) and quantum technologies is most evident in **parametric insurance**. Unlike traditional insurance, where claims are settled after on-the-ground loss adjustment, parametric insurance makes payouts automatically when a pre-defined data threshold (called a *trigger*) is reached.

- **EO-based triggers (Earth Observation):** EO refers to satellite and sensor data that capture conditions on the Earth’s surface. In insurance, it is used to define objective triggers. For example:
 - *Flood coverage* linked to Sentinel-1 radar maps of inundation.
 - *Drought coverage* tied to NDVI (Normalized Difference Vegetation Index), a measure of plant health.
 - *Wildfire coverage* triggered by satellite-detected burn severity.
- **Quantum-enabled pricing:** Quantum computing uses quantum physics to perform calculations that are impractical for classical computers. In insurance, it allows much faster and more precise simulations of risk. For example, *multi-peril contracts* (covering correlated risks like hurricane winds plus flooding) can be priced using quantum-accelerated simulations of joint probability distributions.
- **The challenge of basis risk:** *Basis risk* occurs when the payout trigger does not match the policyholder’s actual loss—for instance, if the index signals a drought but the farmer’s crops remain healthy, or vice versa.

versa. Advances in EO are reducing this problem through:

- Multi-sensor fusion (e.g., combining rainfall, soil moisture, and depth data).
- Higher-resolution satellite grids.
- Integration with IoT (Internet of Things) ground sensors.

The long-term viability of parametric insurance depends on shrinking risk to levels that clients, investors, and regulators find acceptable—be trust and uptake hinge on confidence that payouts reflect real-world l

Quantum Computing: Rewriting the Rules of Risk

If EO clarifies what we see, quantum promises to transform what we calculate.

- **Portfolio optimization:** Structuring reinsurance layers or catastrophe bonds involves billions of combinations. Hybrid algorithms like QAOA (Quantum Approximate Optimization Algorithm) can explore these more efficiently than classical heuristics.
- **Tail-risk simulation:** Monte Carlo methods require millions of trials to price rare “1-in-200-year” events. Quantum Amplitude Estimation offers quadratic speedup, compressing weeks of computation into hours.
- **Post-quantum security:** Quantum also threatens today’s encryption. Regulators now expect insurers to migrate toward NIST’s post-quantum cryptography standards—Kyber (encryption) and Dilithium/SPHINX (signatures)—to safeguard sensitive claims and client data.

Quantum won't replace classical modelling overnight, but pilot project already proving its potential. The shift is conceptual: risk management prepare for a world where quantum engines quietly power the capital insurance.

Convergence: Parametric Insurance

The convergence of EO and quantum is most visible in **parametric insurance** where payouts are triggered by independent indices rather than traditional claims adjustment.

- **EO-based triggers:** Flood covers tied to Sentinel-1 inundation map, drought covers tied to NDVI vegetation indices, wildfire covers triggered by satellite-detected burn severity.
- **Quantum-enabled pricing:** Multi-peril contracts (e.g. hurricane and flood) can be priced more precisely using quantum-accelerated simulations of joint distributions.

The Achilles' heel is **basis risk**: when triggers don't reflect actual losses. Advances—multi-sensor fusion (rainfall + soil moisture + flood depth), higher resolution grids, and integration of IoT sensors—are helping close this gap. Trust and uptake depend on shrinking basis risk to tolerable levels.

Market Context: Capital Flows and Protection Gaps

These innovations are unfolding against a backdrop of surging capital flows and persistent protection gaps.

- **Cat bond boom:** Issuance hit a record \$21 bn in the past 12 months

\$17 bn in H1 2025 alone. Outstanding ILS risk capital surpassed \$55 with average deal sizes up 12%. Parametric cat bonds—often satellite triggered—are gaining market share.

- **Investor demand:** Returns remain robust—Swiss Re's cat bond index posted **14% in 2024** and nearly **50% over five years**. The first catastrophe bond ETF was launched in 2025, expanding retail access.
- **Protection gap pressures:** Insured losses hit ~\$100 bn in H1 2025, yet globally, ~60% of natural catastrophe losses remain uninsured. Even in mature markets, coverage gaps persist. Parametric products are seen as one lever to chip away at these gaps.
- **Fragility of alternative capital:** Hedge funds and ILS investors now over \$115 bn in risk capital, but history shows they can retreat sharply after large events. Innovations like EO-verified triggers and quantified priced parametric structures could strengthen investor confidence by making risks more transparent and measurable.

Governance at the Frontier

Boards and regulators face a dual mandate: harness innovation while ensuring trust.

1. **Model validation:** EO-based indices must be empirically tested against historical loss data, with quantified error rates disclosed.
2. **Customer clarity:** Triggers should be simple and transparent (“pay ≥50% of area flooded”).
3. **Crypto-agility:** Boards must oversee migration to post-quantum standards, with milestone tracking and fallback strategies.

4. **Cross-functional oversight:** Risk, IT, audit, actuarial, and sustainability teams should jointly review EO and quantum deployments to avoid spots.

These governance demands echo a shared theme at **ICML 2025** and **SEEQA 2025**: frontier breakthroughs are inseparable from frontier oversight.

For investors

The rise of Earth Observation and quantum tools in catastrophe insurance signals a market that might be further maturing into greater transparency, speed, and resilience. Parametric products with satellite-based triggers reduce disputes, quantum-enhanced models sharpen pricing, and post-quantum security frameworks strengthen trust—together de-risking capital flows into catastrophe bonds and ILS. With issuance already at record levels and ~60% of global catastrophe losses still uninsured, these innovations expand the investable universe while improving return predictability. Governance will be the differentiator: firms that demonstrate transparent data, crypto-agility, and credible oversight will attract more capital, positioning catastrophe assets to move from niche alternatives toward mainstream portfolio resilience.

Everyday Impact

For customers, this convergence translates into tangible improvement

- **More stable premiums:** Better hazard differentiation reduces overpricing and market withdrawal.
- **Faster payouts:** Flood or wildfire claims can be verified in days, not months.

months—parametric pilots have already paid farmers via mobile m within a week.

- **System resilience:** More accurate models and stronger cryptograp ensure insurers can keep coverage available even under climate st cyber risk.

Closing Reflection

“What do ICML 2025 and SEEQA 2025 have in common?” Both demons that the frontier of technology—whether AI or quantum—is not just ab breakthroughs in algorithms or hardware. It is about building the gove frameworks that allow society to trust these breakthroughs in practice

In catastrophe insurance, the shift is profound: from resignation in the “acts of God” to proactive stewardship of “acts of data.” Satellites sharp what we see, quantum accelerates what we can compute, and governa ensures these tools reinforce resilience rather than amplify fragility.

The fatalism of yesterday is giving way to a new promise: that when disaster strike, data, models, and oversight can make recovery faster, fairer, and reliable.

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